RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105

CS23632 Cryptography and Network Security

Laboratory Record Note Book

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Year / Branch / Section :	3rd Year - CSE - E.
University Register No. :	2116230701275
College Roll No. :	230701275
Semester :	VI Semester
Academic Year :	2025-26

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23632 Cryptography and Network Security

NAME	**SAI ARAVIND S**
ROLL NO	**230701275**
YEAR	**3rd Year - CSE**
SEC	**E**

RAJALAKSHMI ENGINEERING COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: SAI ARAVIND S

Academic Year: 2025-25 **Semester:** VI **Branch:** CSE

Register No. 2116230701275

Certified that this is the bonafide record of work done by the above student in the Cryptography and Network Security Laboratory during the academic year 2026 – 2027.

Signature of Faculty in-charge

Submitted for the Practical Examination held on ……

Internal Examiner External Examiner

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EXPERIMENT – 1

Installation and Configuration of Kali Linux in Oracle VirtualBox

Step 1: Install VirtualBox

Download and install Oracle VirtualBox on your system.

Step 2: Download Kali Linux

Download the ISO file of Kali Linux (64-bit) from the official website.

Step 3: Create Virtual Machine

Open VirtualBox and click “New”.
Set:

- Name: Kali Linux
- Type: Linux
- Version: Debian (64-bit)

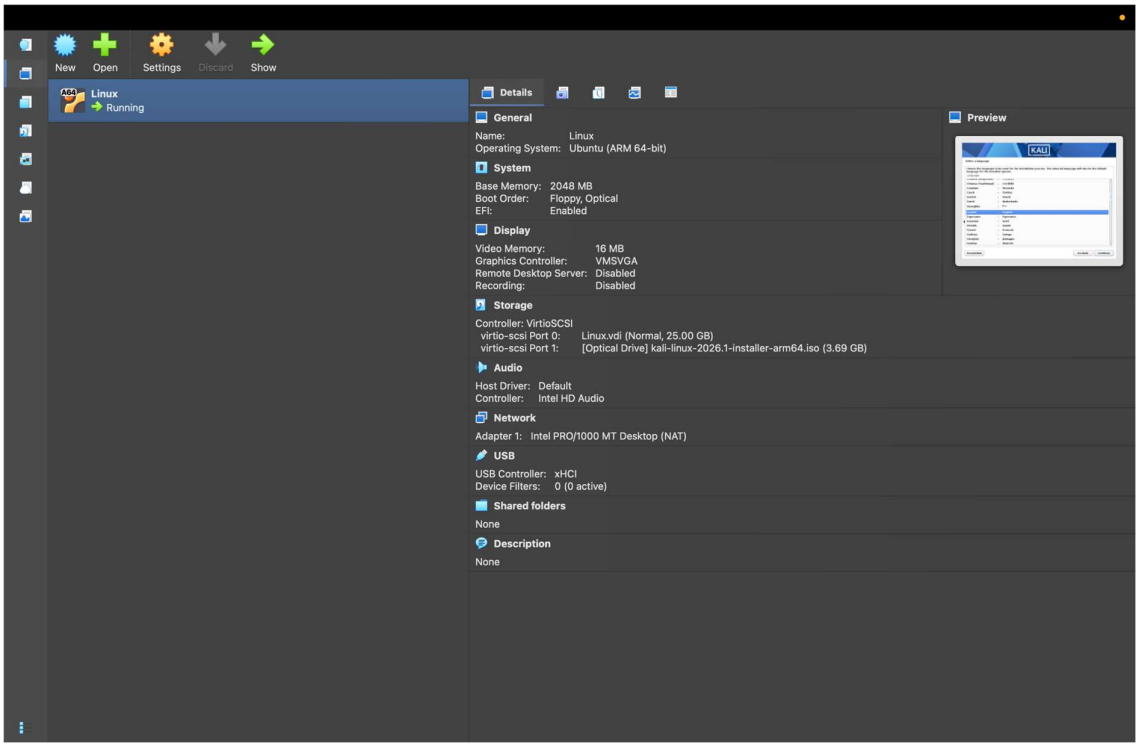
Step 4: Allocate Resources

Assign:

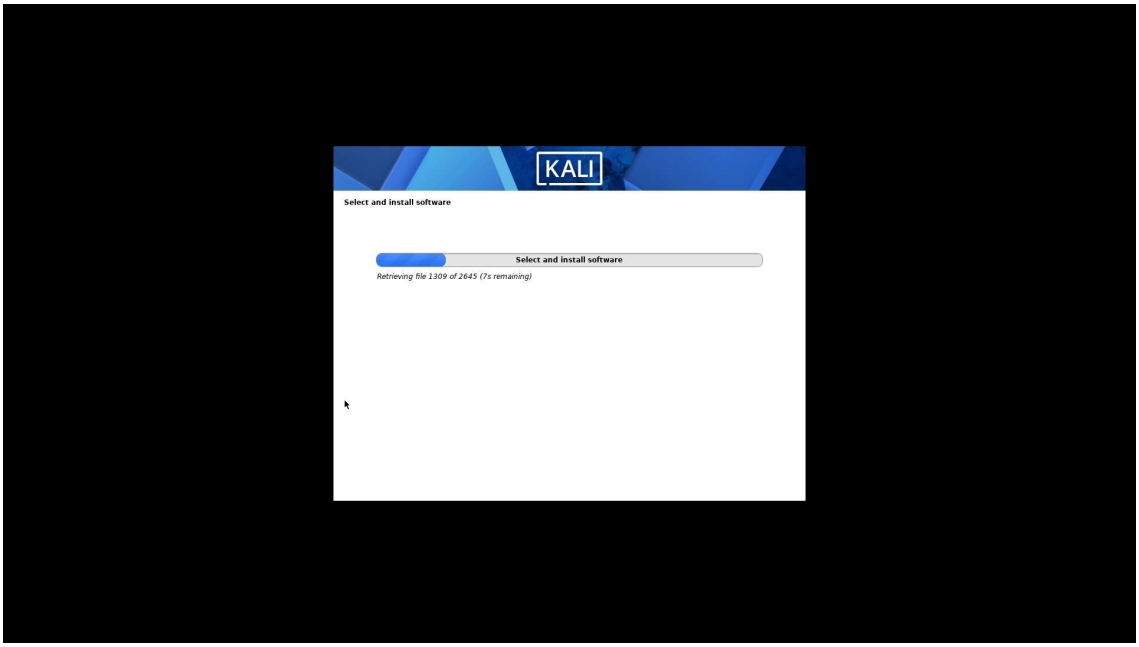
1. RAM: Minimum 2 GB
2. CPU: 2 cores
3. Disk size: 30 GB (VDI, dynamically allocated)

Step 5: Install Kali Linux

Attach the ISO file and start the virtual machine.
Select “Graphical Install” and complete the setup by configuring language, user credentials, and disk partition. Reboot after installation.


New
Open
Settings
Discard
Show
Linux
Running
Details
General
Name: Linux
Operating System: Ubuntu (ARM 64-bit)
System
Base Memory: 2048 MB
Boot Order: Floppy, Optical
EFI: Enabled
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: VirtioSCSI
virtio-scsi Port 0: Linux.vdi (Normal, 25.00 GB)
virtio-scsi Port 1: [Optical Drive] kali-linux-2026.1-installer-arm64.iso (3.69 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: xHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None
Preview


KALI
Select a language
Choose the language to be used for the installation process. The selected language will also be the default language for the installed system.
Language:
Chinese (Traditional) - 中文(繁體)
Croatian - Hrvatski
Czech - Čeština
Danish - Dansk
Dutch - Nederlands
Dzongkha -
English - English
Esperanto - Esperanto
Estonian - Eesti
Finnish - Suomi
French - Français
Galician - Galego
Georgian -
German - Deutsch
Screenshot
Go Back
Continue



Result

Kali Linux was successfully installed and configured in Oracle VirtualBox.

EXPERIMENT – 2

Encryption – Crypto 101 using TryHackMe

Step 1: Login to Platform

Open browser and go to TryHackMe.
Login using your account.

Step 2: Open Crypto 101 Room

Search for **Crypto 101** and click **Join Room**.

Step 3: Start Environment

Click **Start Machine / AttackBox** to launch the virtual lab.

Step 4: Perform Cryptography Tasks

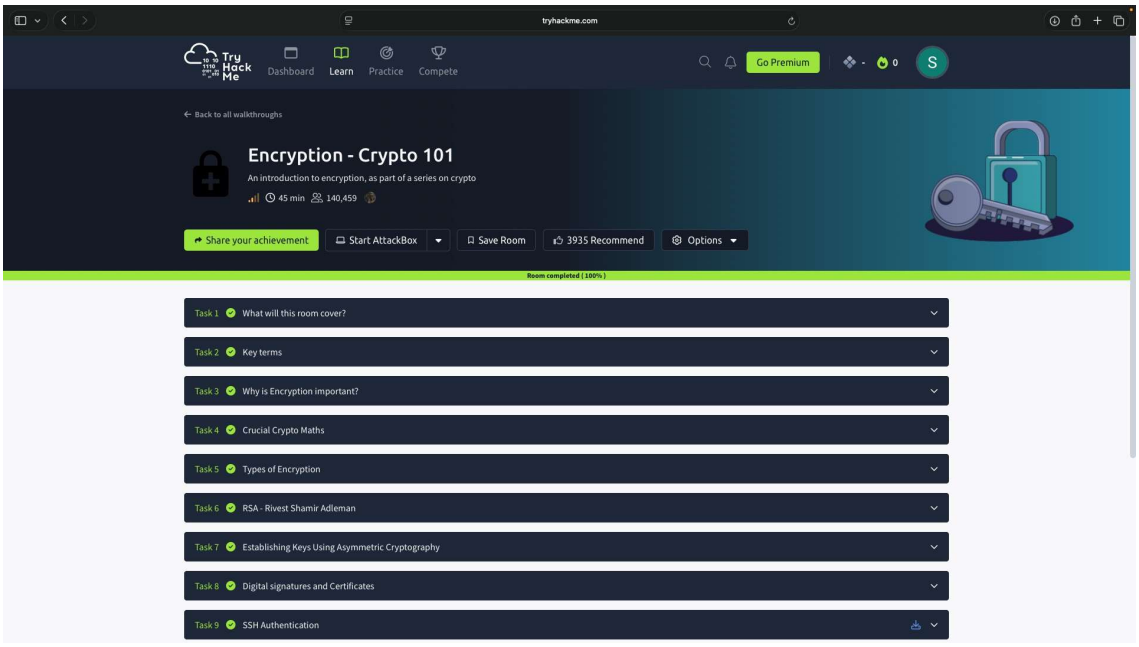
(a) Base64 Encoding/Decoding

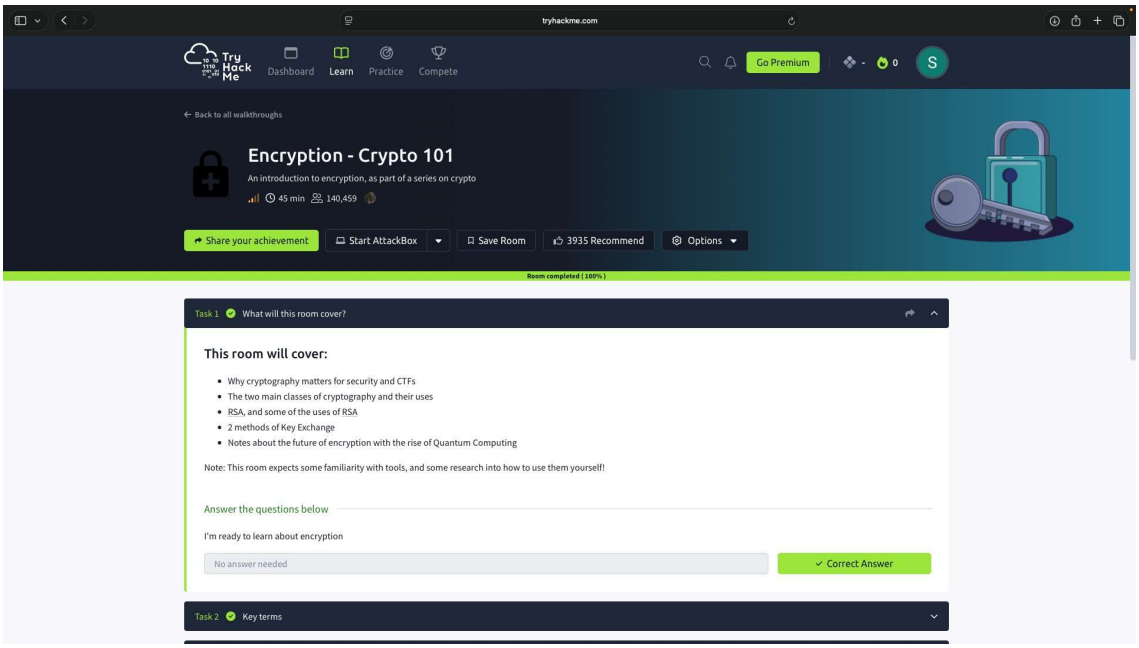
(b) Hex Conversion

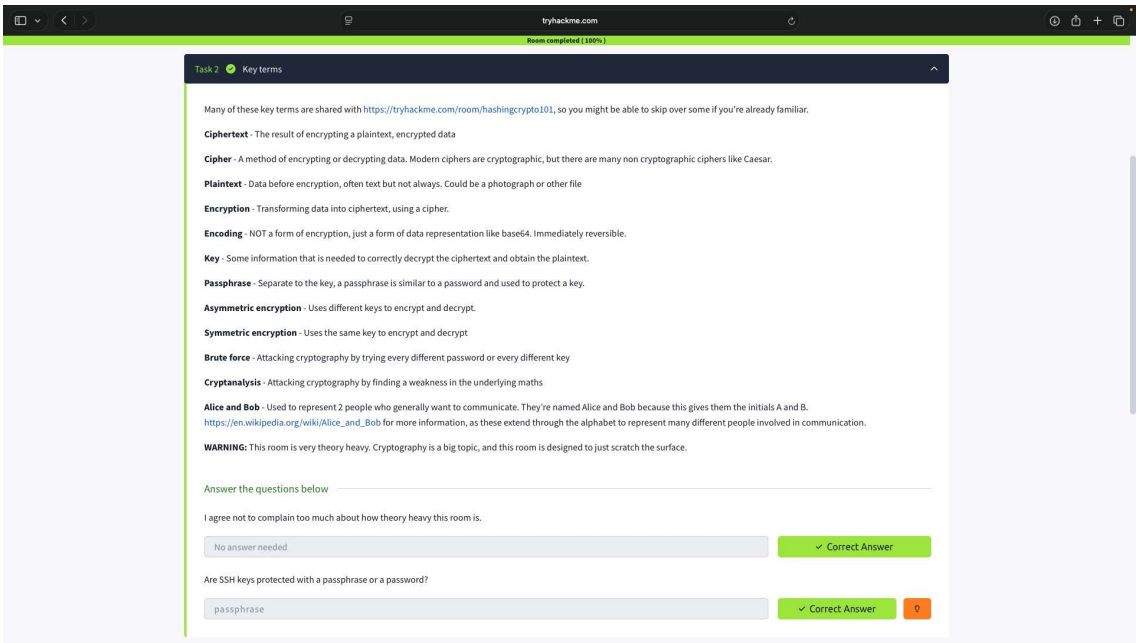
(c) Caesar Cipher

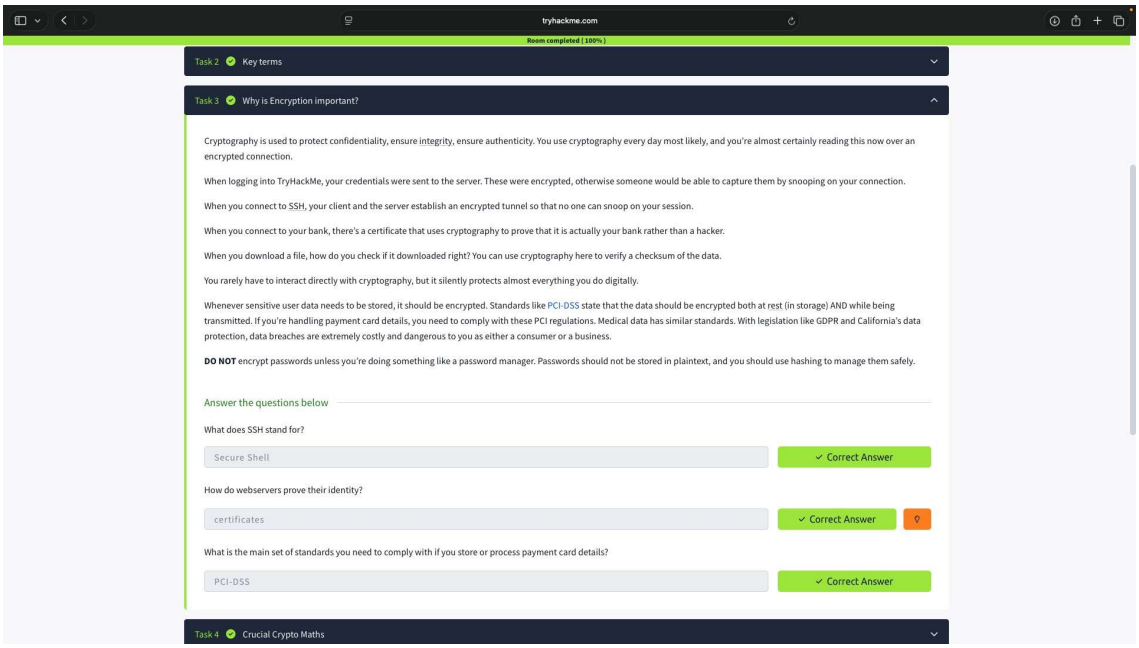
Step 5: Submit Answers

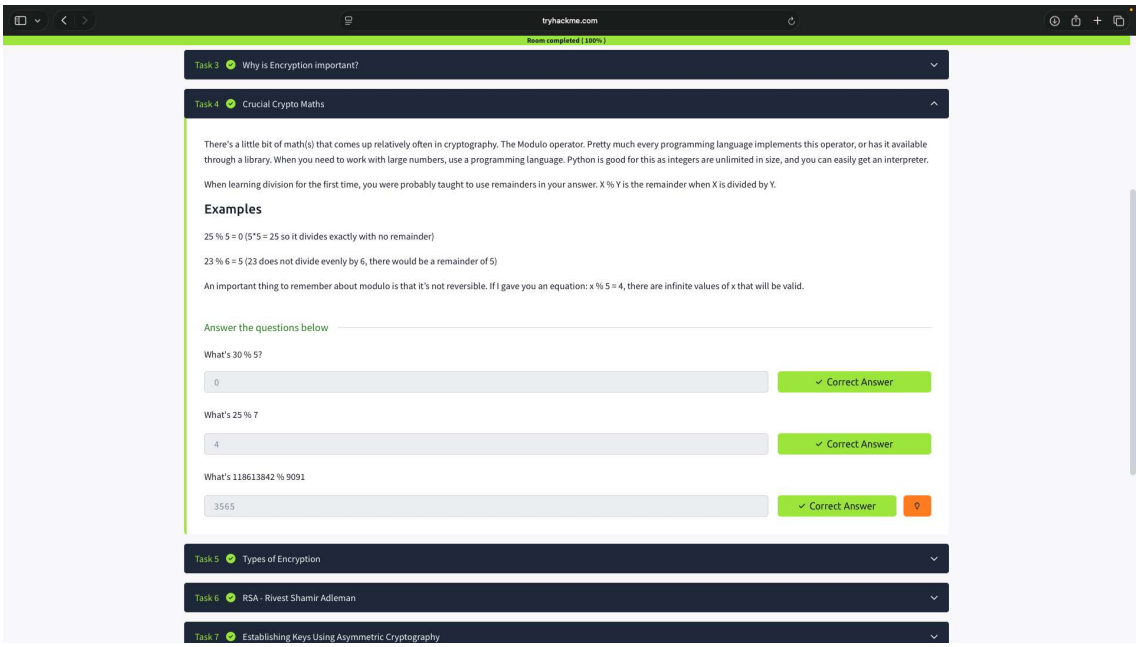
Enter the decoded results in the answer boxes and complete all tasks.

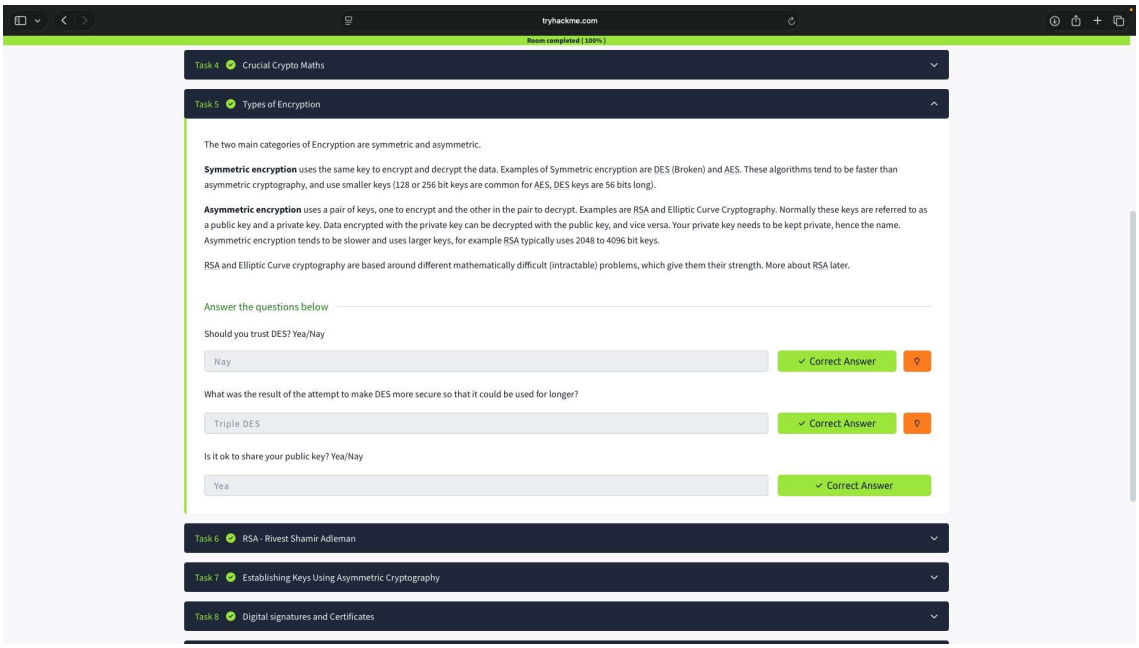

tryhackme.com
Try Hack Me
Dashboard
Learn
Practice
Compete
Go Premium
0
S
Back to all walkthroughs
Encryption - Crypto 101
An introduction to encryption, as part of a series on crypto
45 min
140,459
Share your achievement
Start AttackBox
Save Room
3935 Recommend
Options
Room completed (100%)
Task 1 What will this room cover?
Task 2 Key terms
Task 3 Why is Encryption important?
Task 4 Crucial Crypto Maths
Task 5 Types of Encryption
Task 6 RSA - Rivest Shamir Adleman
Task 7 Establishing Keys Using Asymmetric Cryptography
Task 8 Digital signatures and Certificates
Task 9 SSH Authentication

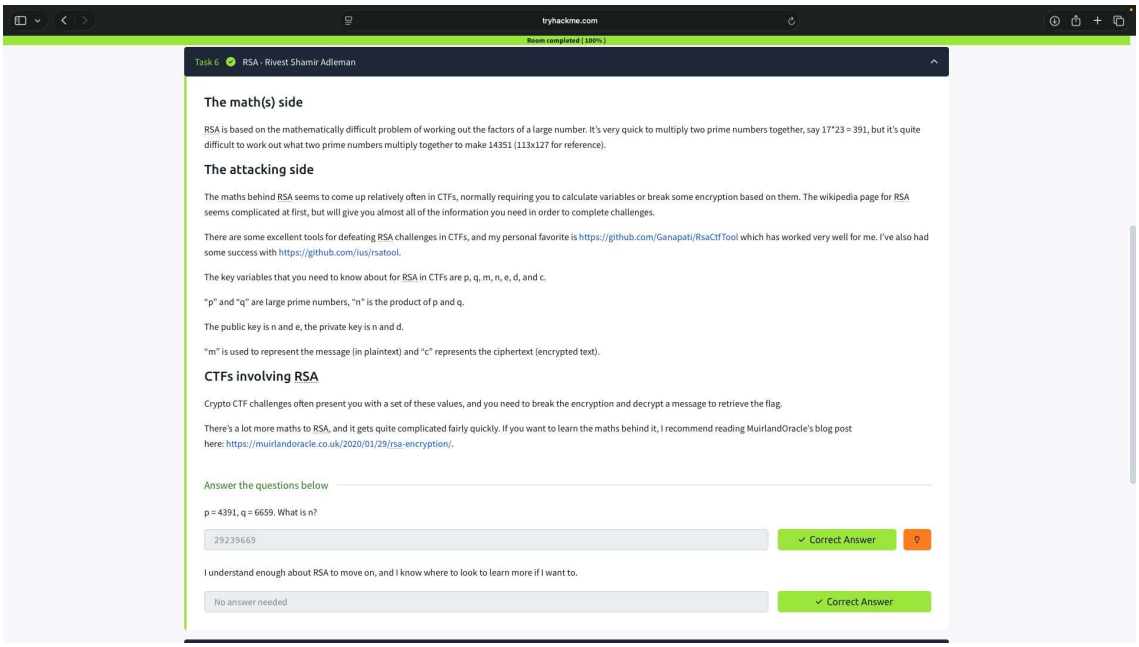

tryhackme.com
Try Hack Me
Dashboard
Learn
Practice
Compete
Go Premium
0
S
Back to all walkthroughs
Encryption - Crypto 101
An introduction to encryption, as part of a series on crypto
45 min
140,459
Share your achievement
Start AttackBox
Save Room
3935 Recommend
Options
Room completed (100%)
Task 1 What will this room cover?
This room will cover:
Why cryptography matters for security and CTFs
The two main classes of cryptography and their uses
RSA, and some of the uses of RSA
2 methods of Key Exchange
Notes about the future of encryption with the rise of Quantum Computing
Note: This room expects some familiarity with tools, and some research into how to use them yourself!
Answer the questions below
I'm ready to learn about encryption
No answer needed
Correct Answer
Task 2 Key terms

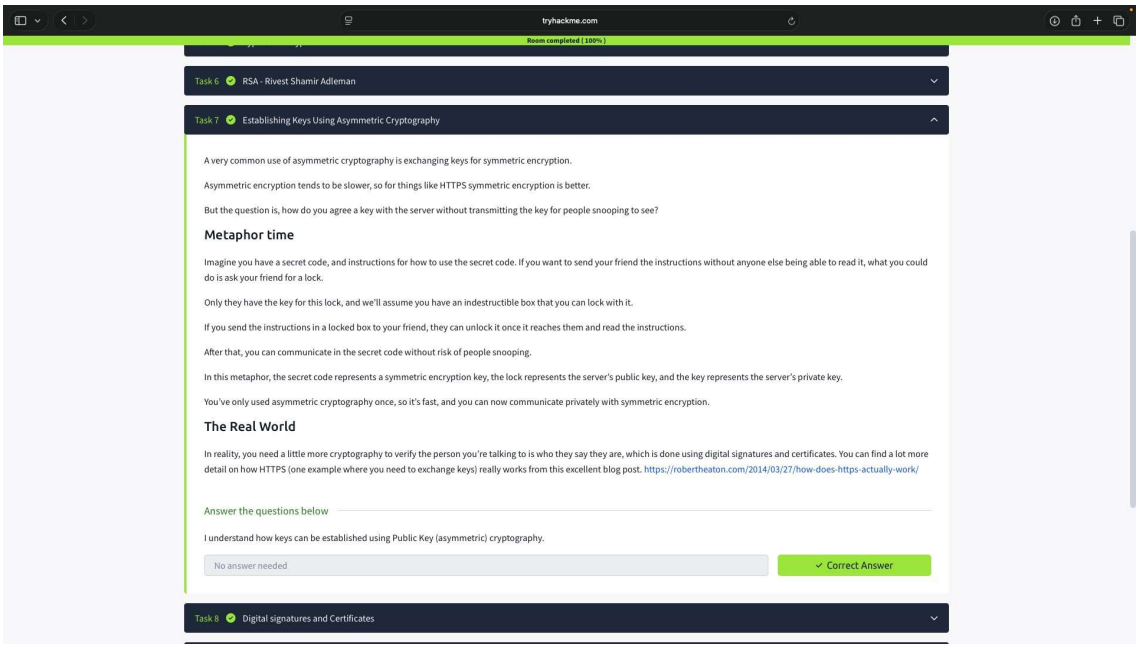

tryhackme.com
Room completed (100%)
Task 2 Key terms
Many of these key terms are shared with https://tryhackme.com/room/hashingcrypto101, so you might be able to skip over some if you're already familiar.
Ciphertext - The result of encrypting a plaintext, encrypted data
Cipher - A method of encrypting or decrypting data. Modern ciphers are cryptographic, but there are many non cryptographic ciphers like Caesar.
Plaintext - Data before encryption, often text but not always. Could be a photograph or other file
Encryption - Transforming data into ciphertext, using a cipher.
Encoding - NOT a form of encryption, just a form of data representation like base64. Immediately reversible.
Key - Some information that is needed to correctly decrypt the ciphertext and obtain the plaintext.
Passphrase - Separate to the key, a passphrase is similar to a password and used to protect a key.
Asymmetric encryption - Uses different keys to encrypt and decrypt.
Symmetric encryption - Uses the same key to encrypt and decrypt
Brute force - Attacking cryptography by trying every different password or every different key
Cryptanalysis - Attacking cryptography by finding a weakness in the underlying maths
Alice and Bob - Used to represent 2 people who generally want to communicate. They're named Alice and Bob because this gives them the initials A and B. https://en.wikipedia.org/wiki/Alice_and_Bob for more information, as these extend through the alphabet to represent many different people involved in communication.
WARNING: This room is very theory heavy. Cryptography is a big topic, and this room is designed to just scratch the surface.
Answer the questions below
I agree not to complain too much about how theory heavy this room is.
No answer needed
Correct Answer
Are SSH keys protected with a passphrase or a password?
passphrase
Correct Answer

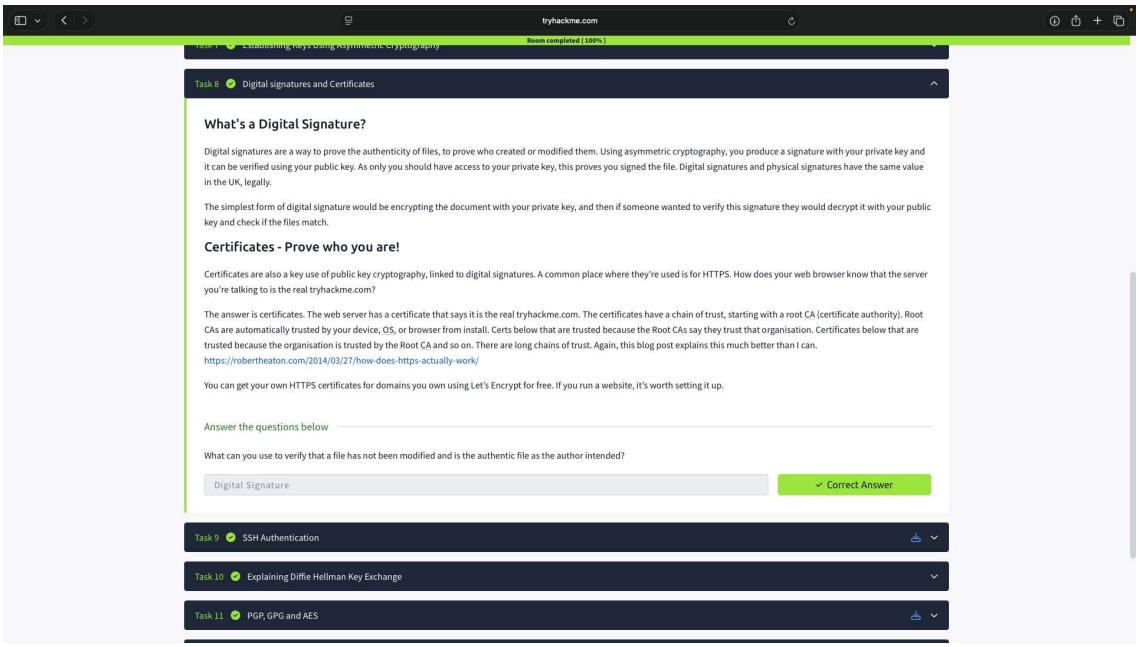

tryhackme.com
Room completed (100%)
Task 2 Key terms
Task 3 Why is Encryption important?
Cryptography is used to protect confidentiality, ensure integrity, ensure authenticity. You use cryptography every day most likely, and you're almost certainly reading this now over an encrypted connection.
When logging into TryHackMe, your credentials were sent to the server. These were encrypted, otherwise someone would be able to capture them by snooping on your connection.
When you connect to SSH, your client and the server establish an encrypted tunnel so that no one can snoop on your session.
When you connect to your bank, there's a certificate that uses cryptography to prove that it is actually your bank rather than a hacker.
When you download a file, how do you check if it downloaded right? You can use cryptography here to verify a checksum of the data.
You rarely have to interact directly with cryptography, but it silently protects almost everything you do digitally.
Whenever sensitive user data needs to be stored, it should be encrypted. Standards like PCI-DSS state that the data should be encrypted both at rest (in storage) AND while being transmitted. If you're handling payment card details, you need to comply with these PCI regulations. Medical data has similar standards. With legislation like GDPR and California's data protection, data breaches are extremely costly and dangerous to you as either a consumer or a business.
DO NOT encrypt passwords unless you're doing something like a password manager. Passwords should not be stored in plaintext, and you should use hashing to manage them safely.
Answer the questions below
What does SSH stand for?
Secure Shell
Correct Answer
How do webservers prove their identity?
certificates
Correct Answer
What is the main set of standards you need to comply with if you store or process payment card details?
PCI-DSS
Correct Answer
Task 4 Crucial Crypto Maths

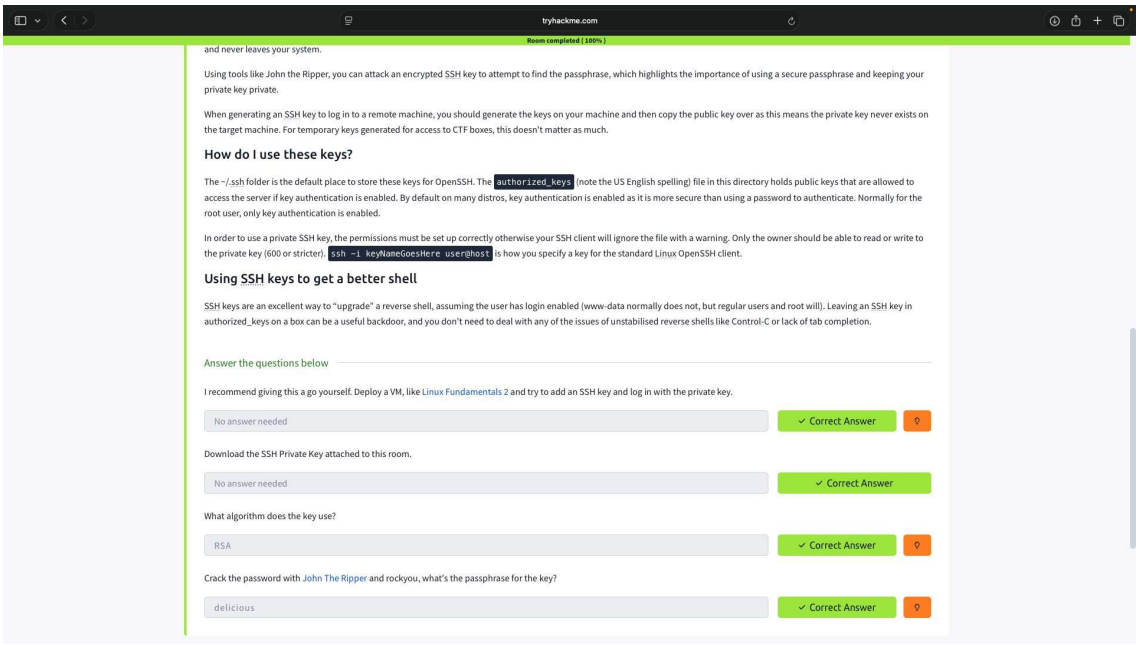

tryhackme.com
Room completed (100%)
Task 3 Why is Encryption important?
Task 4 Crucial Crypto Maths
There's a little bit of math(s) that comes up relatively often in cryptography. The Modulo operator. Pretty much every programming language implements this operator, or has it available through a library. When you need to work with large numbers, use a programming language. Python is good for this as integers are unlimited in size, and you can easily get an interpreter.
When learning division for the first time, you were probably taught to use remainders in your answer. X % Y is the remainder when X is divided by Y.
Examples
25 % 5 = 0 (5*5 = 25 so it divides exactly with no remainder)
23 % 6 = 5 (23 does not divide evenly by 6, there would be a remainder of 5)
An important thing to remember about modulo is that it's not reversible. If I gave you an equation: x % 5 = 4, there are infinite values of x that will be valid.
Answer the questions below
What's 30 % 5?
0
Correct Answer
What's 25 % 7
4
Correct Answer
What's 118613842 % 9091
3565
Correct Answer
Task 5 Types of Encryption
Task 6 RSA - Rivest Shamir Adleman
Task 7 Establishing Keys Using Asymmetric Cryptography

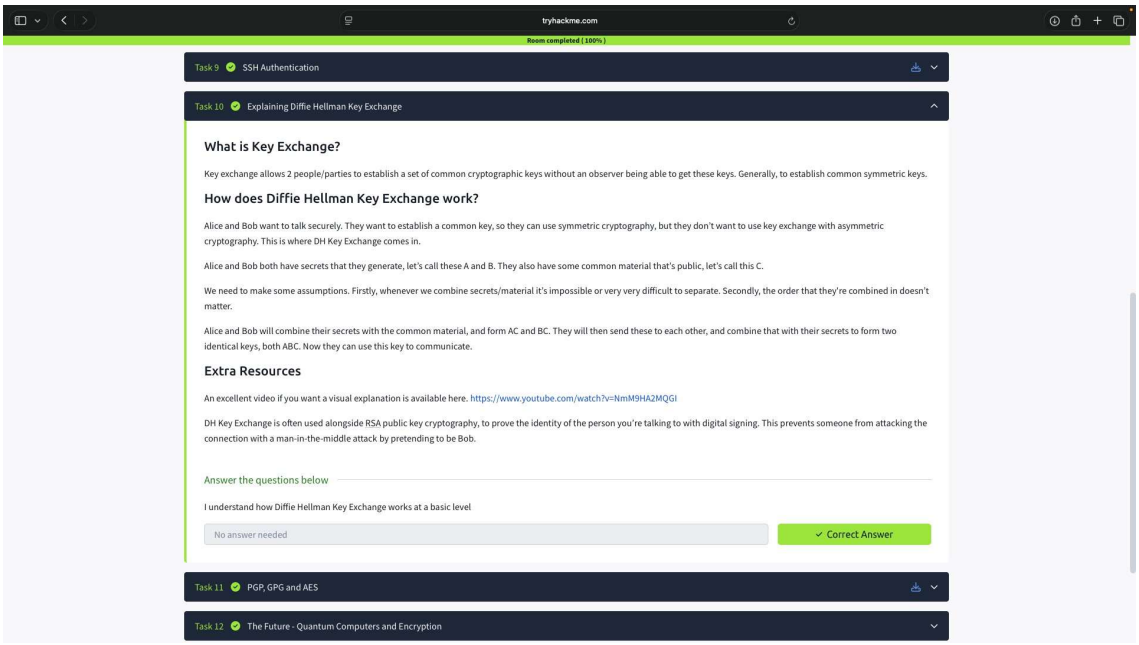

tryhackme.com
Room completed (100%)
Task 4 Crucial Crypto Maths
Task 5 Types of Encryption
The two main categories of Encryption are symmetric and asymmetric.
Symmetric encryption uses the same key to encrypt and decrypt the data. Examples of Symmetric encryption are DES (Broken) and AES. These algorithms tend to be faster than asymmetric cryptography, and use smaller keys (128 or 256 bit keys are common for AES, DES keys are 56 bits long).
Asymmetric encryption uses a pair of keys, one to encrypt and the other in the pair to decrypt. Examples are RSA and Elliptic Curve Cryptography. Normally these keys are referred to as a public key and a private key. Data encrypted with the private key can be decrypted with the public key, and vice versa. Your private key needs to be kept private, hence the name. Asymmetric encryption tends to be slower and uses larger keys, for example RSA typically uses 2048 to 4096 bit keys.
RSA and Elliptic Curve cryptography are based around different mathematically difficult (intractable) problems, which give them their strength. More about RSA later.
Answer the questions below
Should you trust DES? Yea/Nay
Nay
Correct Answer
What was the result of the attempt to make DES more secure so that it could be used for longer?
Triple DES
Correct Answer
Is it ok to share your public key? Yea/Nay
Yea
Correct Answer
Task 6 RSA - Rivest Shamir Adleman
Task 7 Establishing Keys Using Asymmetric Cryptography
Task 8 Digital signatures and Certificates

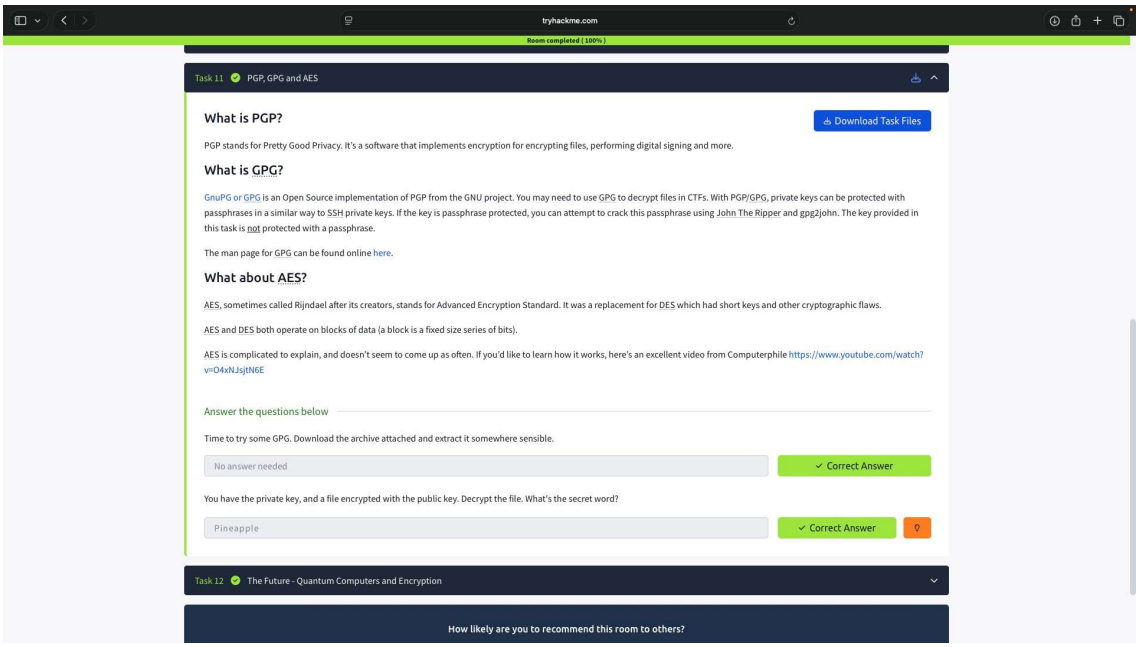

tryhackme.com
Room completed (100%)
Task 6 RSA - Rivest Shamir Adleman
The math(s) side
RSA is based on the mathematically difficult problem of working out the factors of a large number. It's very quick to multiply two prime numbers together, say 17*23 = 391, but it's quite difficult to work out what two prime numbers multiply together to make 14351 (113x127 for reference).
The attacking side
The maths behind RSA seems to come up relatively often in CTFs, normally requiring you to calculate variables or break some encryption based on them. The wikipedia page for RSA seems complicated at first, but will give you almost all of the information you need in order to complete challenges.
There are some excellent tools for defeating RSA challenges in CTFs, and my personal favorite is https://github.com/Ganapati/RsaCtfTool which has worked very well for me. I've also had some success with https://github.com/ius/rsatool.
The key variables that you need to know about for RSA in CTFs are p, q, m, n, e, d, and c.
"p" and "q" are large prime numbers, "n" is the product of p and q.
The public key is n and e, the private key is n and d.
"m" is used to represent the message (in plaintext) and "c" represents the ciphertext (encrypted text).
CTFs involving RSA
Crypto CTF challenges often present you with a set of these values, and you need to break the encryption and decrypt a message to retrieve the flag.
There's a lot more maths to RSA, and it gets quite complicated fairly quickly. If you want to learn the maths behind it, I recommend reading MuirlandOracle's blog post here: https://muirlandoracle.co.uk/2020/01/29/rsa-encryption/.
Answer the questions below
p = 4391, q = 6659. What is n?
29239669
Correct Answer
I understand enough about RSA to move on, and I know where to look to learn more if I want to.
No answer needed
Correct Answer

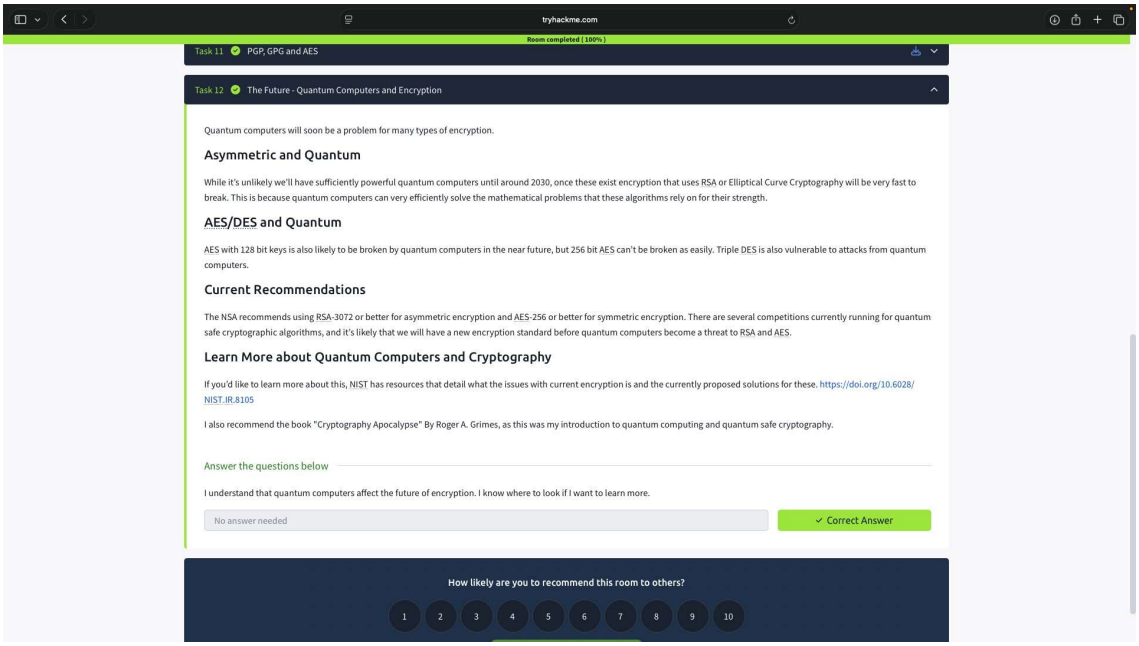

tryhackme.com
Room completed (100%)
Task 6 RSA - Rivest Shamir Adleman
Task 7 Establishing Keys Using Asymmetric Cryptography
A very common use of asymmetric cryptography is exchanging keys for symmetric encryption.
Asymmetric encryption tends to be slower, so for things like HTTPS symmetric encryption is better.
But the question is, how do you agree a key with the server without transmitting the key for people snooping to see?
Metaphor time
Imagine you have a secret code, and instructions for how to use the secret code. If you want to send your friend the instructions without anyone else being able to read it, what you could do is ask your friend for a lock.
Only they have the key for this lock, and we'll assume you have an indestructible box that you can lock with it.
If you send the instructions in a locked box to your friend, they can unlock it once it reaches them and read the instructions.
After that, you can communicate in the secret code without risk of people snooping.
In this metaphor, the secret code represents a symmetric encryption key, the lock represents the server's public key, and the key represents the server's private key.
You've only used asymmetric cryptography once, so it's fast, and you can now communicate privately with symmetric encryption.
The Real World
In reality, you need a little more cryptography to verify the person you're talking to is who they say they are, which is done using digital signatures and certificates. You can find a lot more detail on how HTTPS (one example where you need to exchange keys) really works from this excellent blog post. https://robertheaton.com/2014/03/27/how-does-https-actually-work/
Answer the questions below
I understand how keys can be established using Public Key (asymmetric) cryptography.
No answer needed
Correct Answer
Task 8 Digital signatures and Certificates


tryhackme.com
Room completed (100%)
Task 8 Digital signatures and Certificates
What's a Digital Signature?
Digital signatures are a way to prove the authenticity of files, to prove who created or modified them. Using asymmetric cryptography, you produce a signature with your private key and it can be verified using your public key. As only you should have access to your private key, this proves you signed the file. Digital signatures and physical signatures have the same value in the UK, legally.
The simplest form of digital signature would be encrypting the document with your private key, and then if someone wanted to verify this signature they would decrypt it with your public key and check if the files match.
Certificates - Prove who you are!
Certificates are also a key use of public key cryptography, linked to digital signatures. A common place where they're used is for HTTPS. How does your web browser know that the server you're talking to is the real tryhackme.com?
The answer is certificates. The web server has a certificate that says it is the real tryhackme.com. The certificates have a chain of trust, starting with a root CA (certificate authority). Root CAs are automatically trusted by your device, OS, or browser from install. Certs below that are trusted because the Root CAs say they trust that organisation. Certificates below that are trusted because the organisation is trusted by the Root CA and so on. There are long chains of trust. Again, this blog post explains this much better than I can.
https://robertheaton.com/2014/03/27/how-does-https-actually-work/
You can get your own HTTPS certificates for domains you own using Let's Encrypt for free. If you run a website, it's worth setting it up.
Answer the questions below
What can you use to verify that a file has not been modified and is the authentic file as the author intended?
Digital Signature
✓ Correct Answer
Task 9 SSH Authentication
Task 10 Explaining Diffie Hellman Key Exchange
Task 11 PGP, GPG and AES


tryhackme.com
Room completed (100%)
and never leaves your system.
Using tools like John the Ripper, you can attack an encrypted SSH key to attempt to find the passphrase, which highlights the importance of using a secure passphrase and keeping your private key private.
When generating an SSH key to log in to a remote machine, you should generate the keys on your machine and then copy the public key over as this means the private key never exists on the target machine. For temporary keys generated for access to CTF boxes, this doesn't matter as much.
How do I use these keys?
The ~/.ssh folder is the default place to store these keys for OpenSSH. The authorized_keys (note the US English spelling) file in this directory holds public keys that are allowed to access the server if key authentication is enabled. By default on many distros, key authentication is enabled as it is more secure than using a password to authenticate. Normally for the root user, only key authentication is enabled.
In order to use a private SSH key, the permissions must be set up correctly otherwise your SSH client will ignore the file with a warning. Only the owner should be able to read or write to the private key (600 or stricter). ssh -i keyNameGoesHere user@host is how you specify a key for the standard Linux OpenSSH client.
Using SSH keys to get a better shell
SSH keys are an excellent way to "upgrade" a reverse shell, assuming the user has login enabled (www-data normally does not, but regular users and root will). Leaving an SSH key in authorized_keys on a box can be a useful backdoor, and you don't need to deal with any of the issues of unstabilised reverse shells like Control-C or lack of tab completion.
Answer the questions below
I recommend giving this a go yourself. Deploy a VM, like Linux Fundamentals 2 and try to add an SSH key and log in with the private key.
No answer needed
✓ Correct Answer
Download the SSH Private Key attached to this room.
No answer needed
✓ Correct Answer
What algorithm does the key use?
RSA
✓ Correct Answer
Crack the password with John The Ripper and rockyou, what's the passphrase for the key?
delicious
✓ Correct Answer


tryhackme.com
Room completed (100%)
Task 9 SSH Authentication
Task 10 Explaining Diffie Hellman Key Exchange
What is Key Exchange?
Key exchange allows 2 people/parties to establish a set of common cryptographic keys without an observer being able to get these keys. Generally, to establish common symmetric keys.
How does Diffie Hellman Key Exchange work?
Alice and Bob want to talk securely. They want to establish a common key, so they can use symmetric cryptography, but they don't want to use key exchange with asymmetric cryptography. This is where DH Key Exchange comes in.
Alice and Bob both have secrets that they generate, let's call these A and B. They also have some common material that's public, let's call this C.
We need to make some assumptions. Firstly, whenever we combine secrets/material it's impossible or very very difficult to separate. Secondly, the order that they're combined in doesn't matter.
Alice and Bob will combine their secrets with the common material, and form AC and BC. They will then send these to each other, and combine that with their secrets to form two identical keys, both ABC. Now they can use this key to communicate.
Extra Resources
An excellent video if you want a visual explanation is available here. https://www.youtube.com/watch?v=NmM9HA2MQGI
DH Key Exchange is often used alongside RSA public key cryptography, to prove the identity of the person you're talking to with digital signing. This prevents someone from attacking the connection with a man-in-the-middle attack by pretending to be Bob.
Answer the questions below
I understand how Diffie Hellman Key Exchange works at a basic level
No answer needed
✓ Correct Answer
Task 11 PGP, GPG and AES
Task 12 The Future - Quantum Computers and Encryption





Result

Basic cryptography concepts such as encoding, decoding, hashing, and classical ciphers were successfully performed using TryHackMe.

EXPERIMENT - 3

Performing MITM Attack Using Ettercap

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine.
Login to the system.

Step 2: Check Network

Open terminal and verify IP address:

ifconfig

Ensure the attacker and target systems are in the same network.

Step 3: Launch Tool

Start Ettercap in graphical mode:

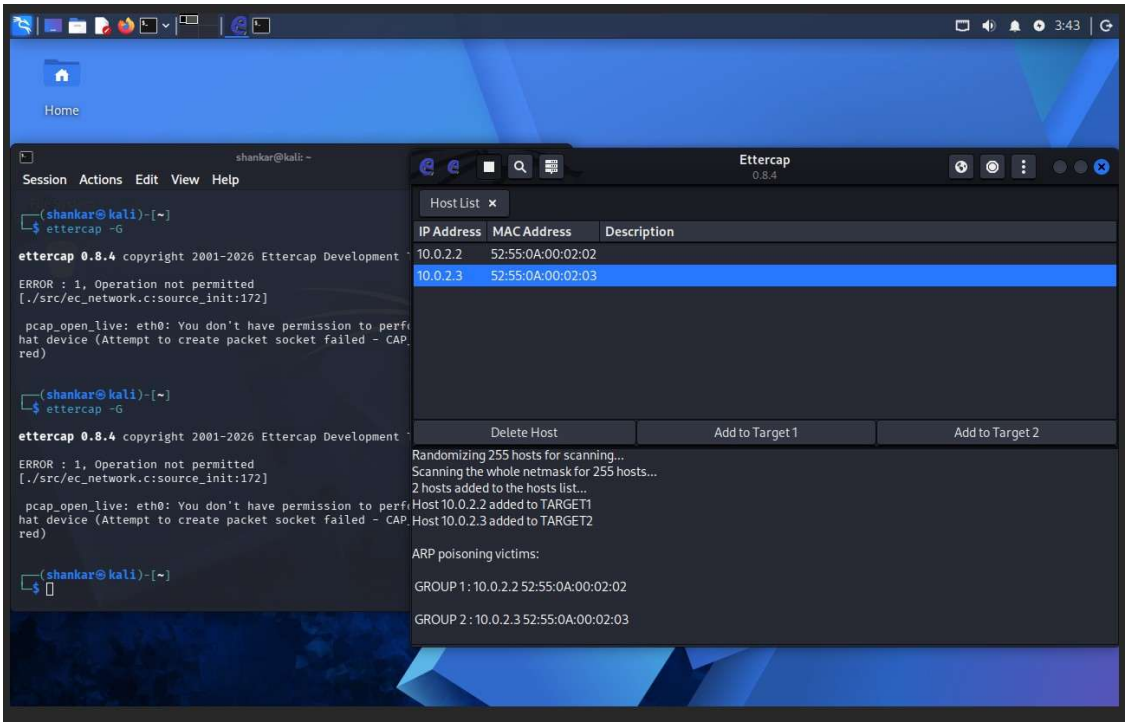
ettercap -G

Step 4: Scan and Select Targets

- Choose **Sniff → Unified Sniffing** and select interface
- Scan for hosts
- Identify:
    1. Target system
    2. Gateway (router)

Step 5: Perform MITM and Monitor

4. Configure MITM using ARP poisoning (conceptually)
5. Start sniffing
6. Observe intercepted network traffic in the tool



Result

The MITM attack was successfully demonstrated in a controlled environment, showing how unsecured communication can be intercepted and why encryption is important.

EXPERIMENT - 4

Hash Cracking Using John the Ripper

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.
Login to the system.

Step 2: Open Terminal and Verify Tool

Open terminal and check if John the Ripper is installed:

```
john --version
```

Step 3: Create Hash File

Create a file and store a password hash:

```
nano hash.txt
```

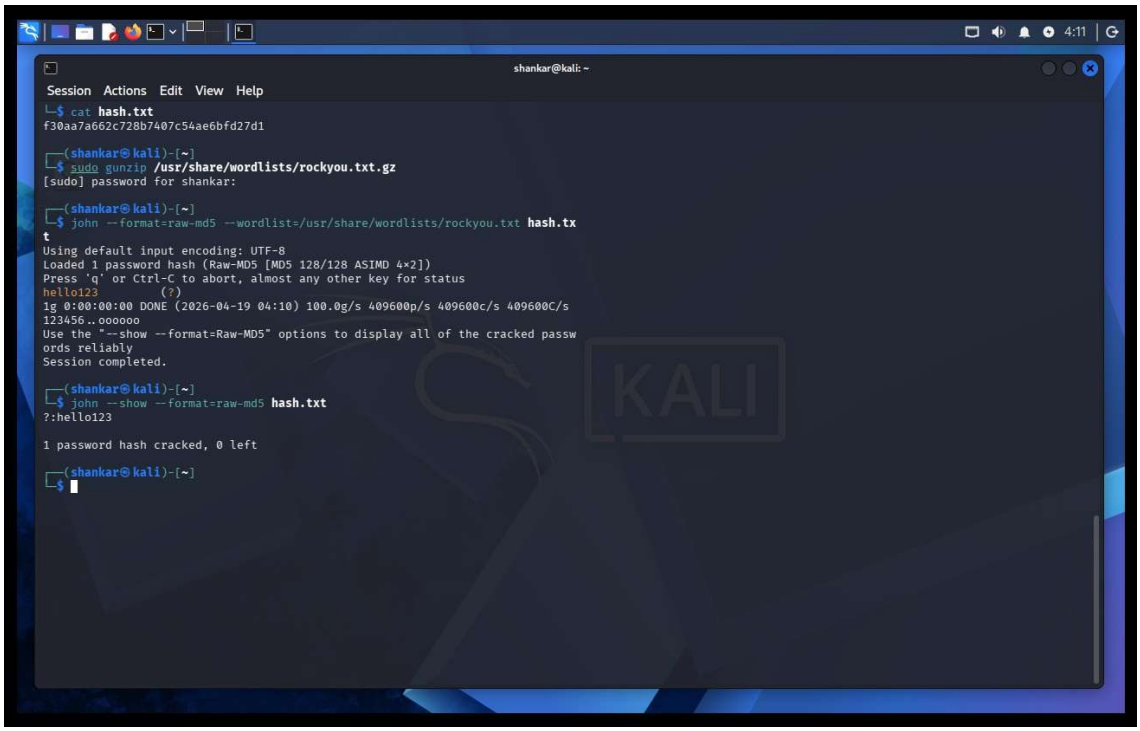
Step 4: Run Hash Cracking

```
john hash.txt
```

The tool attempts to crack the hash using default wordlists.

Step 5: View Cracked Password

```
john --show hash.txt
```



Result

Password hash cracking was successfully demonstrated, highlighting the need for strong passwords and secure hashing techniques.

EXPERIMENT - 5

IPTables Firewall Configuration in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine. Log in to the system.

Step 2: Open Terminal and Check IPTables

Open the terminal and verify that IPTables is installed and available in the system.

Step 3: View Existing Rules

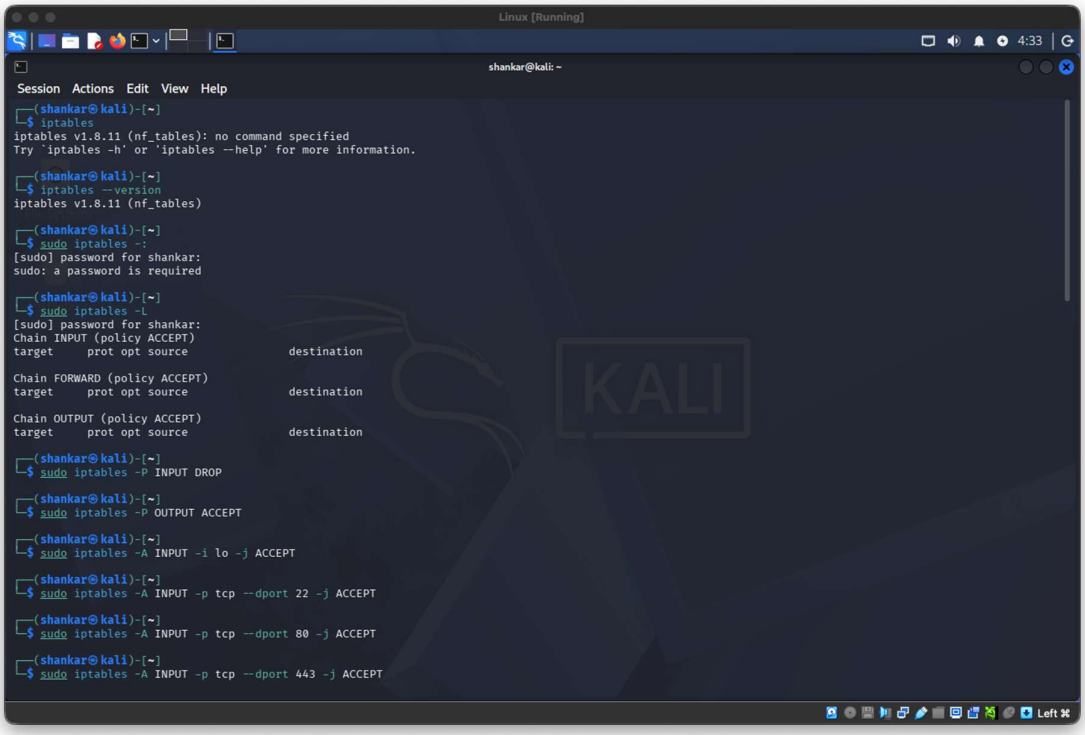
Check the current firewall rules to understand the default configuration before making changes.

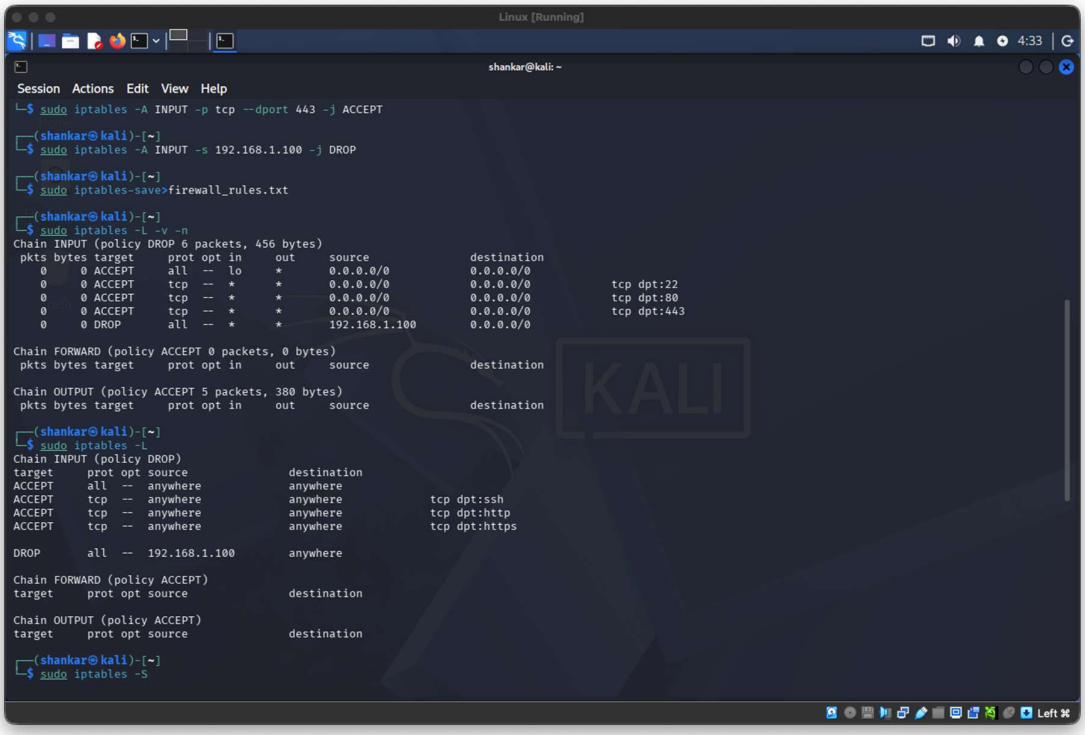
Step 4: Configure Firewall Rules

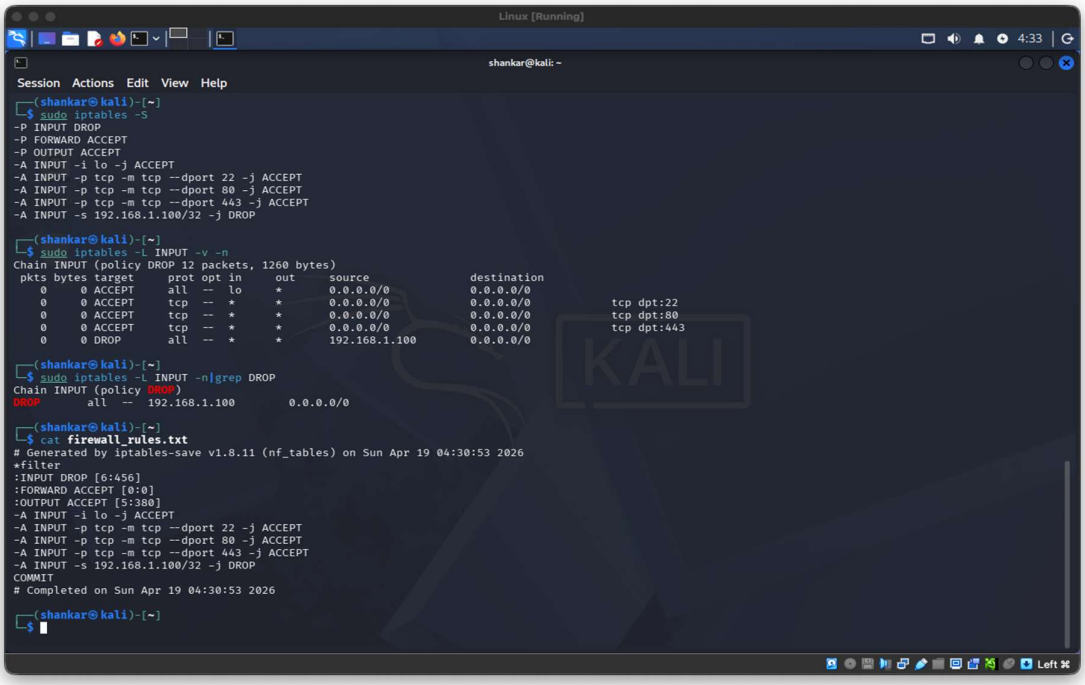
Set default policies to control incoming and outgoing traffic.
Allow necessary services such as localhost communication, SSH, HTTP, and HTTPS.
Block unwanted or suspicious IP addresses.

Step 5: Save and Verify Rules

Save the configured firewall rules and verify that they are applied correctly by reviewing the rule list.


Linux [Running]
4:33
shankar@kali: ~
Session Actions Edit View Help
┌──(shankar㉿kali)-[~]
└─$ iptables
iptables v1.8.11 (nf_tables): no command specified
Try `iptables -h' or 'iptables --help' for more information.
┌──(shankar㉿kali)-[~]
└─$ iptables --version
iptables v1.8.11 (nf_tables)
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -:
[sudo] password for shankar:
sudo: a password is required
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -L
[sudo] password for shankar:
Chain INPUT (policy ACCEPT)
target prot opt source destination
Chain FORWARD (policy ACCEPT)
target prot opt source destination
Chain OUTPUT (policy ACCEPT)
target prot opt source destination
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -P INPUT DROP
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -P OUTPUT ACCEPT
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -i lo -j ACCEPT
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
Left ⌘





Result

Various firewall rules were implemented using IPTables, demonstrating how network traffic can be controlled to enhance system security.

EXPERIMENT - 6

Snort IDS/IPS Using TryHackMe

Step 1: Login to Platform

Open TryHackMe in a browser and log in using your account.

Step 2: Open Snort Room

Search for a Snort-related room and join it to begin the experiment.

Step 3: Start AttackBox

Launch the AttackBox, which provides a ready-to-use Linux environment for performing the tasks.

Step 4: Verify Snort Installation

Open the terminal and confirm that Snort is installed and ready to use.

Step 5: Run Snort in IDS Mode

Start Snort in detection mode so that it monitors network traffic and generates alerts for suspicious activity.

Step 6: Detect Network Threats

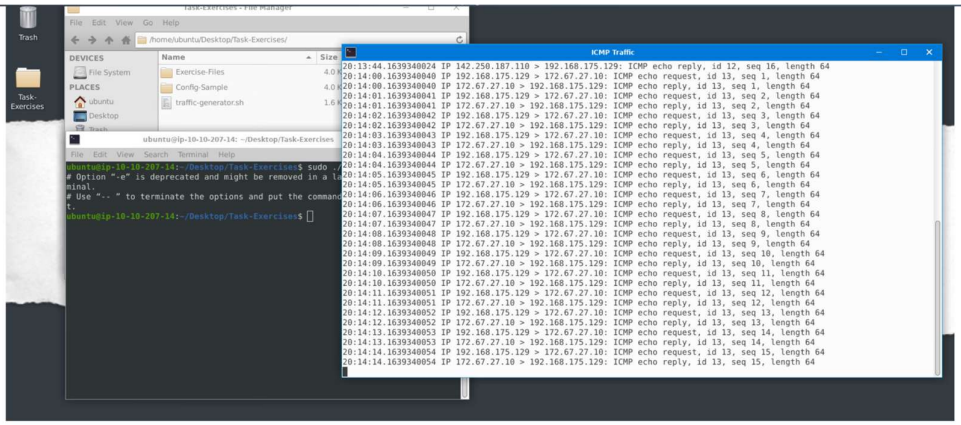
Perform the tasks given in the room and observe how Snort identifies and reports potential threats in real time.

Step 7: Configure IPS Mode

Enable prevention mode to allow Snort to block malicious traffic based on predefined rules.

Step 8: Analyze Alerts

Review the alerts generated by Snort and identify different types of network attacks.



Answer the questions below

Navigate to the Task-Exercises folder and run the command "./.easy.sh" and write the output

Too Easy! ✓ Correct Answer

Task 3 Introduction to IDS/IPS

Task 4 First Interaction with Snort

Once we use a configuration file, Snort has much more power! The configuration file is Snort's all-in-one management file. Rules, plugins, detection mechanisms, default actions, and output settings are identified here. Multiple configuration files are possible for different purposes and cases, but we can only use one at runtime.

Note that Snort automatically shows the default banner and initial information about your setup when you start it. You can prevent this by using the "**-q**" parameter.

Parameter	Description
-V / --version	This parameter provides information about your instance version.
-c	Identifying the configuration file
-T	Snort's self-test parameter allows you to test your setup with this parameter.
-q	Quiet mode prevents Snort from displaying the default banner and initial information about your setup.

That was easy; let's move on to the next task and continue exploring snort modes.

Answer the questions below

Run the Snort instance and check the build number.

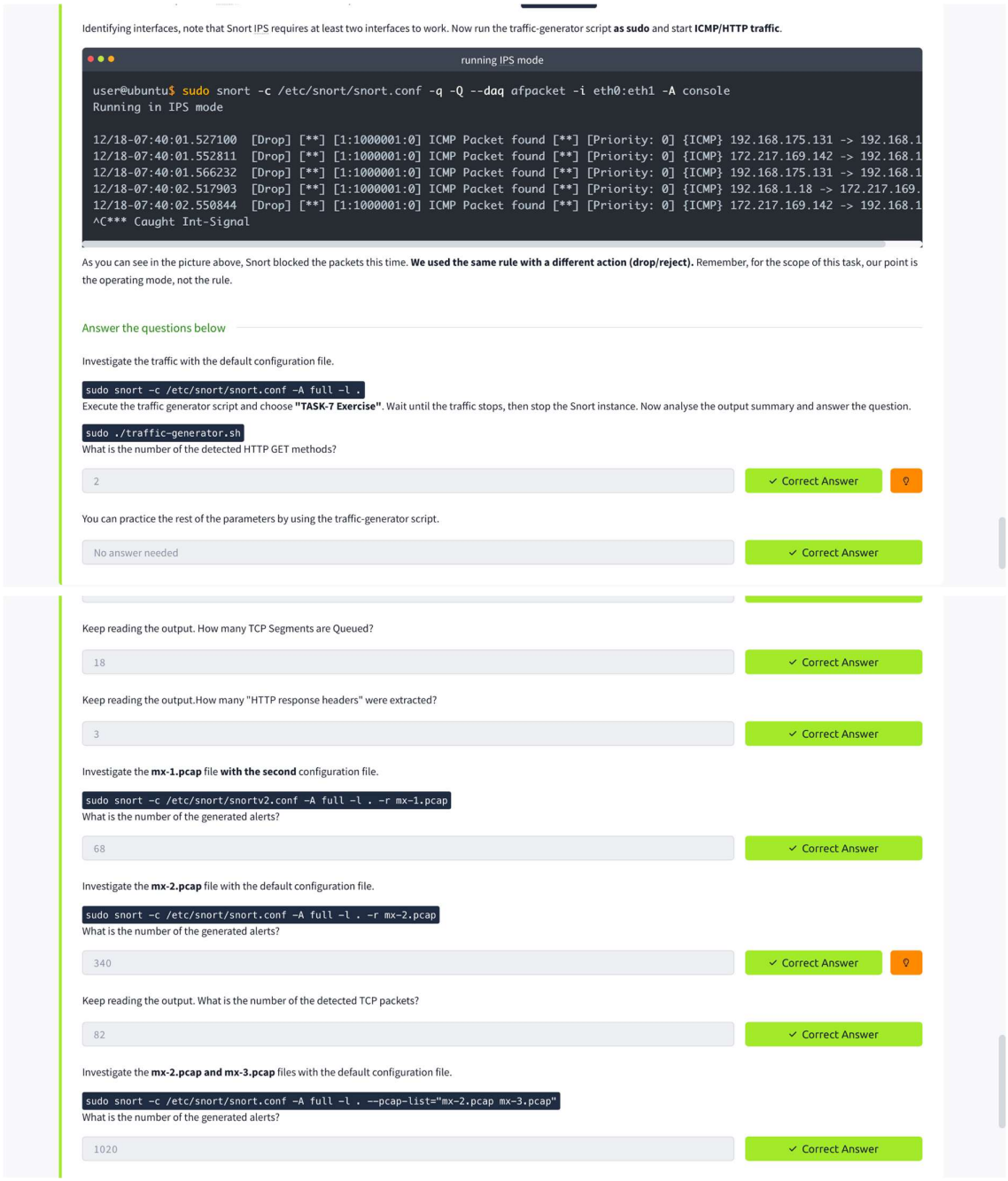
149 ✓ Correct Answer

Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

4151 ✓ Correct Answer

Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

1 ✓ Correct Answer



Result

Snort was successfully used as an IDS/IPS to detect and prevent network threats, demonstrating real-time network security monitoring in a controlled environment.

EXPERIMENT – 7

Performing Code Injection on Application Process Using Ptrace in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine. Log in to the system.

Step 2: Create Target Program

Create a simple C program that runs continuously. This program will act as the target process for testing.

Step 3: Run Target Application

Execute the target program and identify its Process ID (PID), which is required to interact with the running process.

Step 4: Create Injector Program

Write a C program that uses the ptrace system call to attach to the target process and access its internal state such as registers or memory.

Step 5: Compile Programs

Compile both the target and injector programs using a C compiler so they can be executed.

Step 6: Attach to Target Process

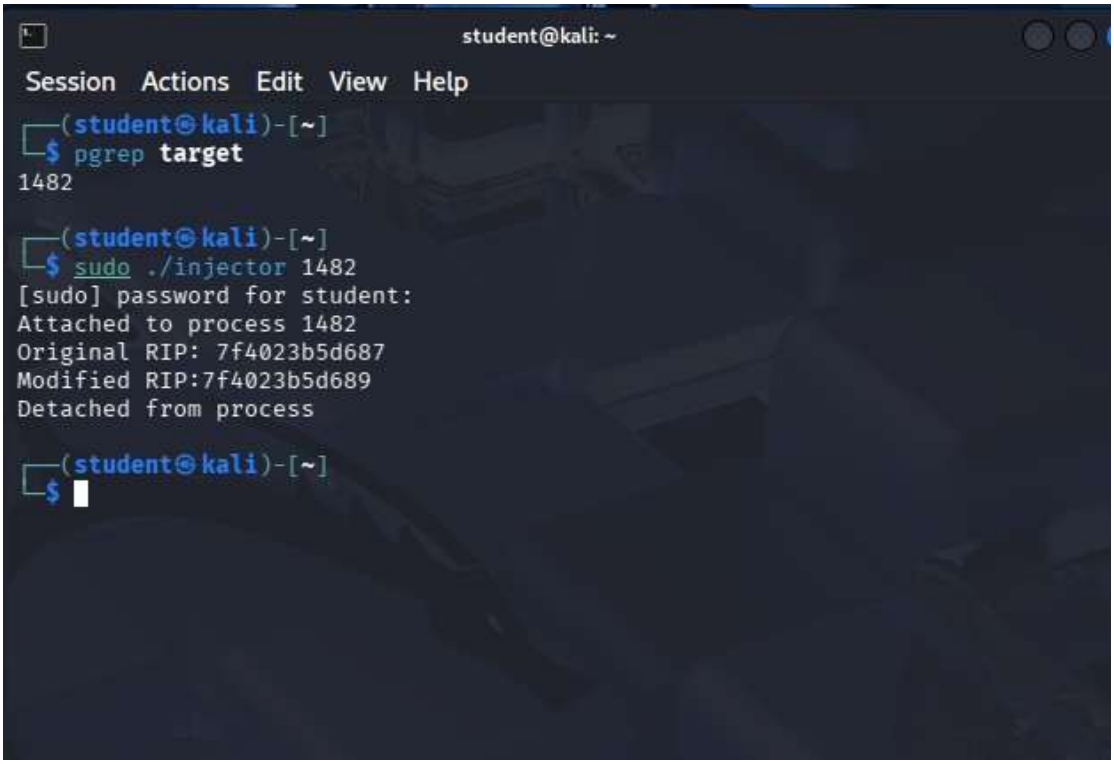
Run the injector program and connect it to the target process using the PID. This allows control over the target process.

Step 7: Modify Execution

Change the target process behavior by altering its execution flow or memory content using ptrace.

Step 8: Observe Changes

Resume the process and observe how its behavior is modified due to the injection.



Result

Code injection using ptrace was successfully demonstrated, showing how process execution can be controlled and highlighting the importance of system-level security and process isolation.

EXPERIMENT - 8

Privilege Escalation Using TryHackMe

Step 1: Login to Platform

Open TryHackMe and log in

Step 2: Open Room

Search for **Linux Privilege Escalation** and join the room.

Step 3: Start AttackBox

Launch the AttackBox and open the terminal.

Step 4: Perform Enumeration

Check user details, system information, and permissions to identify possible vulnerabilities.

Step 5: Identify Vulnerabilities

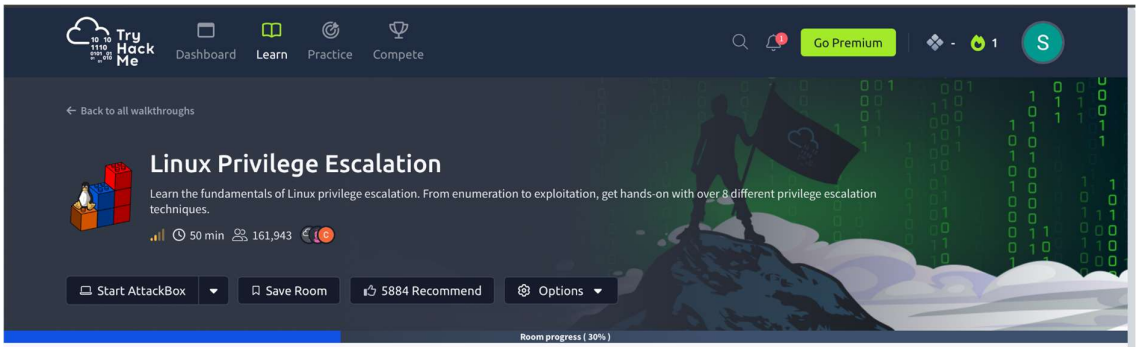
Look for misconfigurations such as sudo permissions, SUID files, weak file access, or cron jobs.

Step 6: Escalate Privileges

Exploit the identified vulnerability to gain higher privileges (root access).

Step 7: Verify and Complete Tasks

Confirm root access and capture required flags to complete the exercise.



Task 1 Introduction

Task 2 What is Privilege Escalation?

Task 3 Enumeration

Task 4 Automated Enumeration Tools

Task 5 Privilege Escalation: Kernel Exploits

Task 6 Privilege Escalation: Sudo

Below is a short example used to find files that have the SUID bit set. The SUID bit allows the file to run with the privilege level of the account that owns it, rather than the account which runs it. This allows for an interesting privilege escalation path,we will see in more details on task 6. The example below is given to complete the subject on the "find" command.

- `find / -perm -u=s -type f 2>/dev/null`: Find files with the SUID bit, which allows us to run the file with a higher privilege level than the current user.

General Linux Commands

As we are in the Linux realm, familiarity with Linux commands, in general, will be very useful. Please spend some time getting comfortable with commands such as `find`, `locate`, `grep`, `cut`, `sort`, etc.

Answer the questions below

What is the hostname of the target system?

wade7363 ✓ Correct Answer

What is the Linux kernel version of the target system?

3.13.0-24-generic ✓ Correct Answer

What Linux is this?

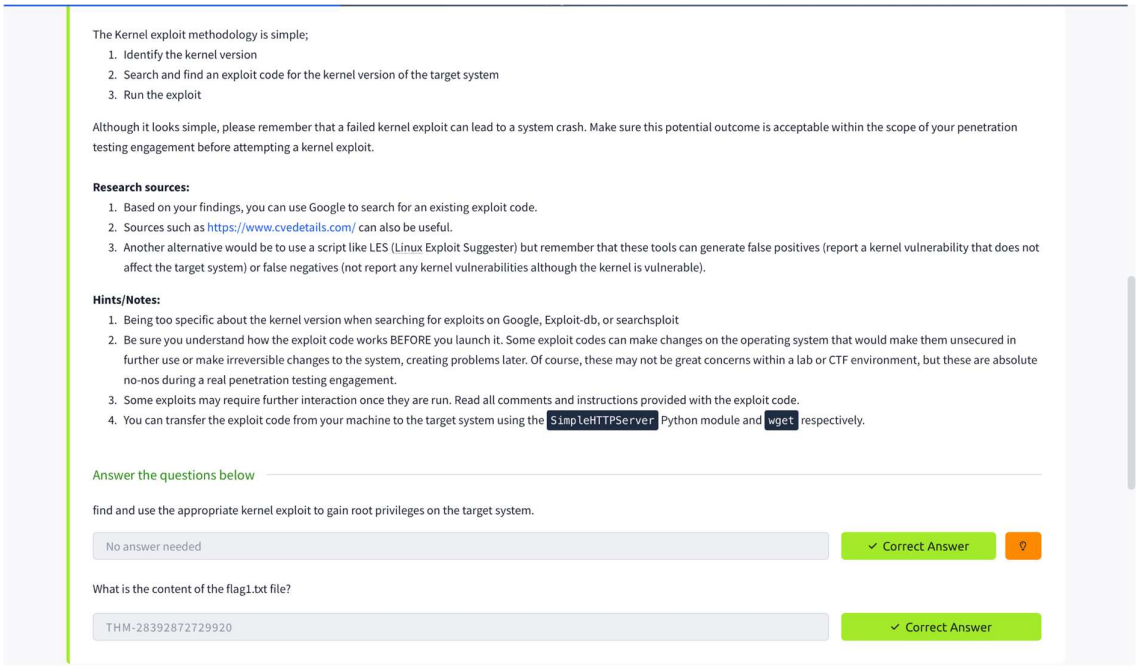
Ubuntu 14.04 LTS ✓ Correct Answer

What version of the Python language is installed on the system?

2.7.6 ✓ Correct Answer

What vulnerability seem to affect the kernel of the target system? (Enter a CVE number)

CVE-2015-1328 ✓ Correct Answer



Result

Privilege escalation was successfully performed, demonstrating how system misconfigurations can lead to unauthorized access and highlighting the importance of secure system practices.

EXPERIMENT - 9

Windows Exploitation Using Metasploit

Step 1: Login to Platform

Open TryHackMe and log in.

Step 2: Open Room

Search for a **Metasploit / Windows Exploitation** room and join it.

Step 3: Start AttackBox

Launch the AttackBox and open the terminal.

Step 4: Launch Metasploit

Start the Metasploit Framework console to begin the exploitation process.

Step 5: Scan Target System

Identify the target system and perform basic scanning to find vulnerabilities.

Step 6: Select Exploit

Choose an appropriate Windows exploit module based on the identified vulnerability.

Step 7: Configure Parameters

Set target details and required options such as payload and connection settings.

Step 8: Execute Exploit

Run the exploit to gain access to the target system.

Step 9: Post-Exploitation

Verify access, check system information, and perform basic enumeration.

Step 10: Complete Tasks

Capture required flags and mark all tasks as completed.

Result

Windows vulnerabilities were successfully exploited using the Metasploit Framework, demonstrating practical knowledge of system exploitation and post-exploitation techniques.

EXPERIMENT – 10

Wireless Audit Using Aircrack-ng

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux. Connect the external Wi-Fi adapter.

Step 2: Verify Wireless Interface

Check the available wireless interface and ensure it is detected properly.

Step 3: Enable Monitor Mode

Switch the wireless adapter to monitor mode to capture network packets.

Step 4: Scan Wireless Networks

Scan nearby Wi-Fi networks and identify the target network details such as BSSID and channel.

Step 5: Capture Handshake

Capture the WPA/WPA2 handshake by monitoring the target network traffic.

Step 6: Perform Password Cracking

Use a wordlist to attempt password recovery from the captured handshake.

Step 7: Stop Monitoring

Disable monitor mode after completing the analysis.

Output

WPA/WPA2 handshake is captured and the password cracking process is performed successfully.

Result

Wireless auditing was successfully demonstrated using Aircrack-ng, highlighting the importance of strong passwords and secure wireless configurations.

EXPERIMENT-11

IP Spoofing Using GitHub Tool in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Prepare Tool

Download or clone an IP spoofing tool from GitHub and install required dependencies.

Step 3: Configure Parameters

Set the target IP address, spoofed source IP, and network interface in the tool.

Step 4: Execute Tool

Run the tool to send packets with a forged source IP address.

Step 5: Monitor Traffic

Observe the network traffic using monitoring tools to verify spoofed packets.

Output

Packets are transmitted with modified (spoofed) source IP addresses and reflected in network traffic.

Result

IP spoofing was successfully demonstrated, showing how packet headers can be manipulated and emphasizing the need for strong network security measures.

EXPERIMENT - 12

Camera Phishing Study in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Study Tool

Refer to an open-source repository on GitHub and understand how camera phishing works.

Step 3: Analyze Workflow

Study how fake permission prompts are created and how users are tricked into granting camera access.

Step 4: Simulated Demonstration

Perform a consent-based simulation and observe camera permission requests and user responses.

Step 5: Identify Risks and Prevention

Understand risks like privacy leakage and learn prevention methods such as permission control and user awareness.

Output

Understanding of camera phishing techniques and associated privacy risks.

Camera Phishing

A page to take pictures from the front and back camera

This program requires php, ngrok and python! Please install these prerequisites before running

requires

- Install Ngrok
  - Widnows: https://ngrok.com/download/windows
  - Linux: https://ngrok.com/download/linux
  - Termux: https://github.com/Yisus7u7/termux-ngrok
- Install php
  - Windows: https://www.php.net/downloads.php
  - Linux: sudo apt install php
  - Termux: pkg install php
- Install Python (py)
  - Windows: Download
  - Linux: sudo apt install python
  - Termux: pkg install python

ScreenShot

After installing ngrok, enter the site and register, get your token and enter the token with this command: `ngrok config add-authtoken <token>`

Now run the program:

```
git clone https://github.com/Mr-Spect3r/CamPhish
cd CamPhish
python main.py
```

Result

Camera phishing concepts were successfully studied, highlighting the importance of user awareness and secure permission handling.

EXPERIMENT - 13

Phishing Attack Study Using Z-Phisher (Simple Steps)

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Study Tool

Refer to the Z-Phisher framework on GitHub and understand its features.

Step 3: Analyze Workflow

Study how phishing pages imitate real websites and trick users into entering credentials.

Step 4: Simulated Demonstration

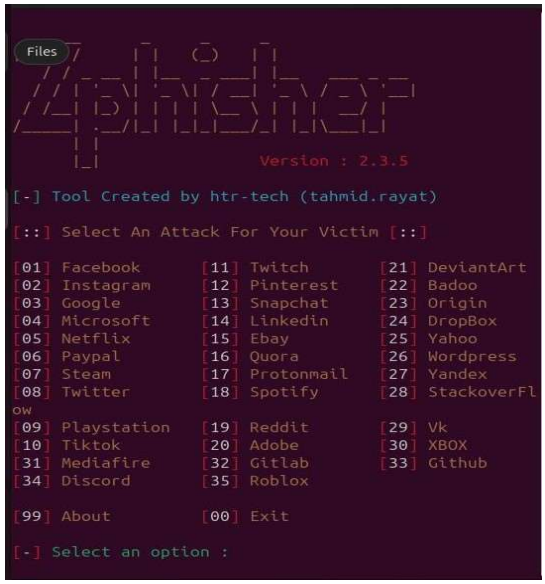
Observe phishing page behavior using a safe, consent-based simulation with dummy data.

Step 5: Identify Risks and Prevention

Understand risks like credential theft and learn prevention methods such as user awareness, HTTPS verification, and two-factor authentication.

Output

Understanding of phishing techniques and social-engineering risks.



Result

Phishing concepts were successfully studied, emphasizing user awareness and security practices.

EXPERIMENT - 14

Brute Force Attack Study in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Setup Test Environment

Use a local test account with dummy credentials in an isolated environment.

Step 3: Study Attack Concept

Understand how repeated login attempts are used to guess passwords.

Step 4: Simulated Demonstration

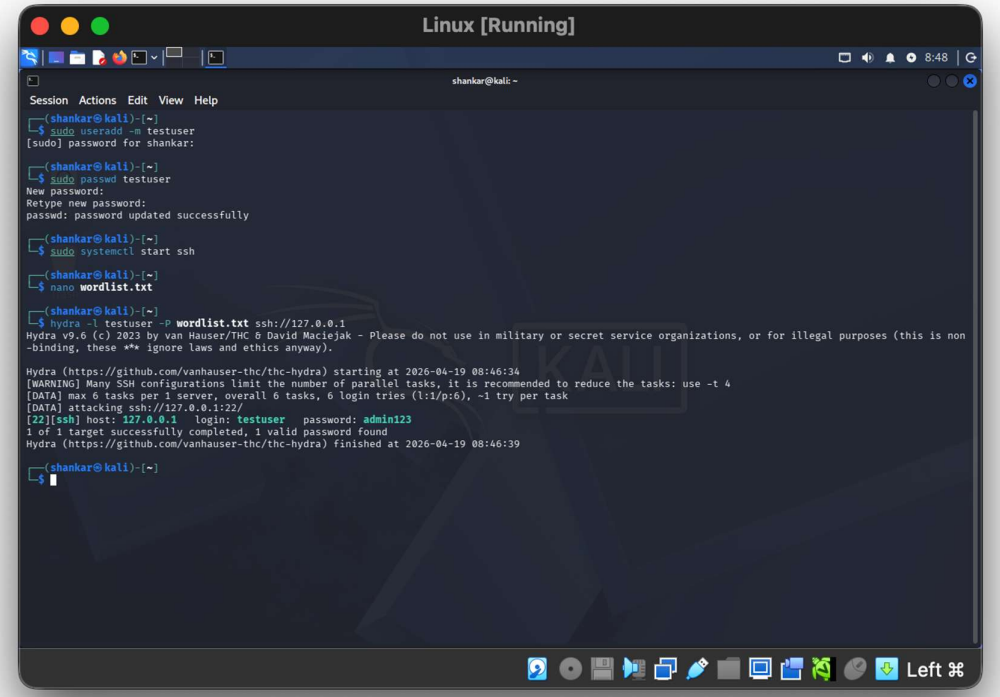
Observe multiple authentication attempts and system responses in a controlled setup.

Step 5: Analyze Risks and Prevention

Identify risks of weak passwords and study prevention methods like strong passwords, account lockout, and multi-factor authentication.

Output

Simulated login attempts demonstrate how weak passwords can be guessed.



Result

Brute force attack concepts were successfully studied, highlighting the importance of strong authentication mechanisms.